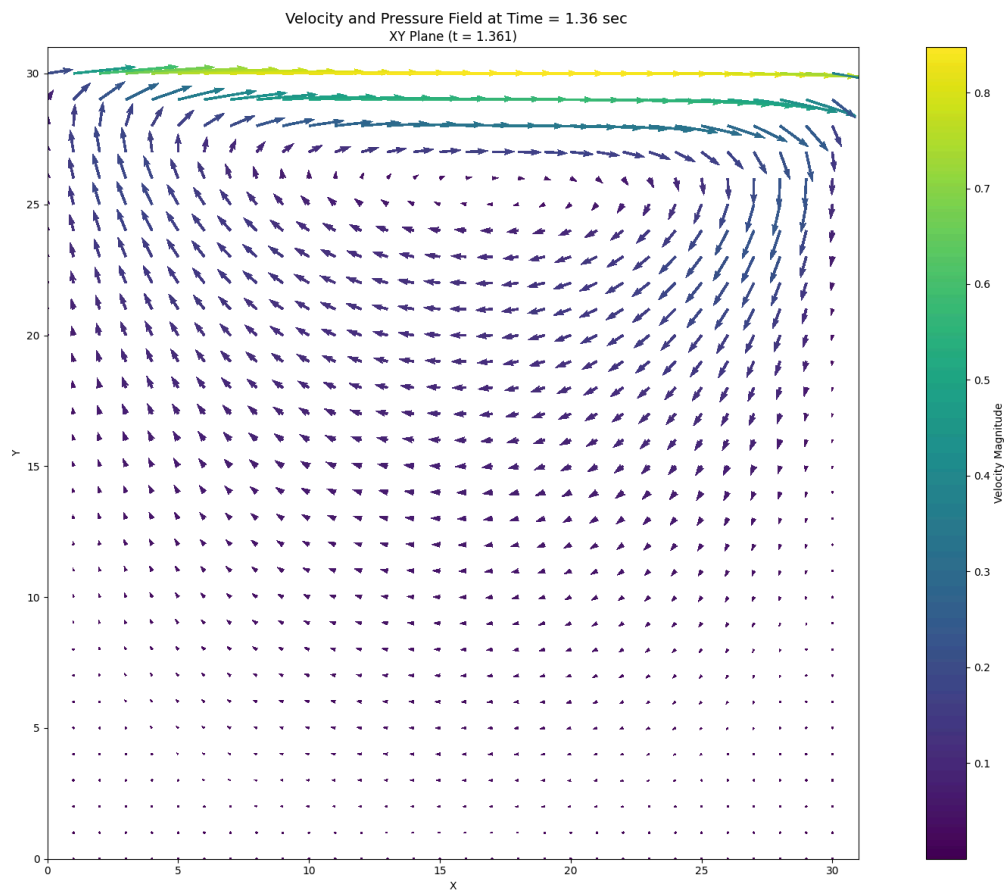


Experimental Report

Case: Lid Driven Cavity

Description of Flow: This case models the standard two-dimensional lid-driven cavity, in which the top wall moves tangentially while the remaining three walls remain stationary. The imposed lid motion generates a primary circulating vortex and weaker secondary recirculation regions, creating a strongly nonlinear velocity field. The experiment aims to recover the streamwise momentum equation from the recorded velocity and pressure data.

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

$$v_t = -1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e-02*v_{xx} + 1.00000000e-02*v_{yy} - 1.00000000e-03*p_y$$

Learned equation

$$u_t = -1.13842289e+00 \cdot uu_x - 1.03055812e+00 \cdot vu_y + 8.40649079e-03 \cdot u_{xx} + 1.03461437e-02 \cdot u_{yy} - 1.04792514e-03 \cdot p_x$$

$$v_t = -1.06616627e+00 \cdot uv_x - 1.25928856e+00 \cdot vv_y + 3.54865766e-03 \cdot v_{xx} + 1.23468066e-02 \cdot v_{yy} - 1.05083164e-03 \cdot p_y$$

Term	Reference coefficient	Learned coefficient	Absolute error
uu_x	-1.00000000e+00	-1.13842289e+00	1.38422889e-01
vu_y	-1.00000000e+00	-1.03055812e+00	3.05581219e-02
u_xx	+1.00000000e-02	+8.40649079e-03	1.59350921e-03
u_yy	+1.00000000e-02	+1.03461437e-02	3.46143665e-04
p_x	-1.00000000e-03	-1.04792514e-03	4.79251383e-05
Term	Reference coefficient	Learned coefficient	Absolute error
uv_x	-1.00000000e+00	-1.06616627e+00	6.61662724e-02
vv_y	-1.00000000e+00	-1.25928856e+00	2.59288563e-01
v_xx	+1.00000000e-02	+3.54865766e-03	6.45134234e-03
v_yy	+1.00000000e-02	+1.23468066e-02	2.34680658e-03
p_y	-1.00000000e-03	-1.05083164e-03	5.08316415e-05

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	rho	1.00000000e+03	kg m ^{-3}
Dynamic viscosity	viscosity	1.00000000e+01	Pa s
Kinematic viscosity	nu	1.00000000e-02	m ² s ^{-1}
Characteristic velocity	u_max	1.00000000e+00	m s ^{-1}
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s ^{-2}
Grid points in x	Nx	3.10000000e+01	cells
Grid points in y	Ny	3.10000000e+01	cells
Grid spacing in x	dx	3.22580645e-02	m
Grid spacing in y	dy	3.22580645e-02	m
Time step	dt	1.00000000e-04	s
Numerical threshold	epsilon	7.00000000e-11	-

3. STRidge Selection and Error Metrics

| Regularisation parameter, λ | 1.00000000e-09 | | L0 penalty | 1.00000000e-06 |

Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	4.35889502e-03
Validation RMSE	6.52899072e-03
Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	1.10494359e-02
Validation RMSE	6.16076275e-03

4. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(6655, 31, 30)
Regression field shape (t, x, y)	(4255, 22, 27)
Regression samples	2,527,470
Dictionary terms	5
First saved time index	150
Temporal crop at each boundary	1200 layers
x-boundary crop	4 points
y-boundary crop	2 points